

Combine overlap correction and back-to-back regression methods to analyse eye-tracking-EEG co-registration data during natural reading

<MSc> Thesis Proposal - <Yanhong Xu>
March 2026

1. Introduction

1.1. Motivation

Natural reading is a complicated cognitive process involving numerous causal and non-causal factors that may be partially correlated and unfold across at visual, attentional, lexical-semantic, and oculomotor levels. Co-registration of eye movements (EMs) and electroencephalography (EEG) enables the investigation of fixation-related potentials (FRPs) during natural reading. A key challenge in FRP data is that rapid fixations induce strong temporal overlap, while lexical factors are often highly collinear. Disentangling the respective contributions of such correlated factors (e.g., word surprisal, word frequency, and word length) from multivariate EEG signals therefore remains a central methodological challenge.

King et al. (2020) introduced back-to-back regression (B2B), a linear regression framework designed to disentangle covarying predictors in multidimensional neural data and to obtain interpretable factor-wise estimates. Yet in EMs-EEG co-registration, rapid event sequences elicit temporally overlapping responses, which can bias coefficient estimates and further complicate inference. A principled pipeline that jointly addresses temporal overlap and predictor collinearity is therefore needed, and it remains unclear how robust and interpretable factor estimates can be obtained in practice.

This project evaluates whether overlap correction using linear deconvolution within a regression event-related potential framework (rERP), implemented in Unfold (Ehinger & Dimigen, 2019), improves B2B's ability to disentangle collinear factors in fixation-locked EEG during natural reading, using simulations with known ground truth and a real-data application to *Zurich Cognitive Language Processing Corpus* (ZuCo) (Hollenstein et al., 2018). To the best of our knowledge, this combination has not been systematically benchmarked in the literature. This work also aims to support more interpretable analyses of NLP-derived lexical predictors (e.g., surprisal) in FRPs, potentially informing both linguistic and cognitive perspectives on natural reading.

1.2. Other work

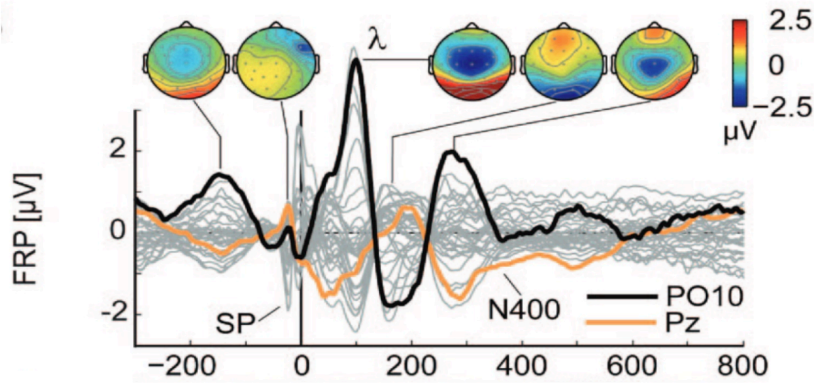
ZuCo dataset provides co-registered eye-tracking and EEG data during natural reading, making it a suitable benchmark for fixation-locked analyses with lexical and oculomotor predictors. In this project, ZuCo is used as a real-data application. Together with overlap correction and B2B, the relevant background is summarised below.

1.2.1. Temporal overlap in FRPs

FRPs are event-related EEG responses time-locked to the onset of eye fixations, with FRP data, it is possible to investigate neural correlates of specific and different aspects of oculomotor behaviour, while also providing insight into the processes underlying reading. For instance, as shown in Figure 1, reading-related EEG responses include well-established components such as the N400, which is

commonly associated with late semantic integration and is often modulated by lexical variables such as word frequency and predictability, as well as earlier visual components linked to occipital processing. (Degno & Liversedge, 2020; Dimigen et al., 2011).

An important issue in co-registration is how to separate temporally overlapping neural components in the EEG. Because multiple covariates can modulate the EEG signal and confound event-related estimates, Ehinger & Dimigen (2019) developed the Unfold toolbox, which implements linear deconvolution-based rERP, enabling less biased estimation of event-related effects and more reliable interpretation.



Source from Dimigen et al. (2011)

Figure 1: **grand average FRP in natural reading.** λ : early visual component. N400: lexico-semantic response component.

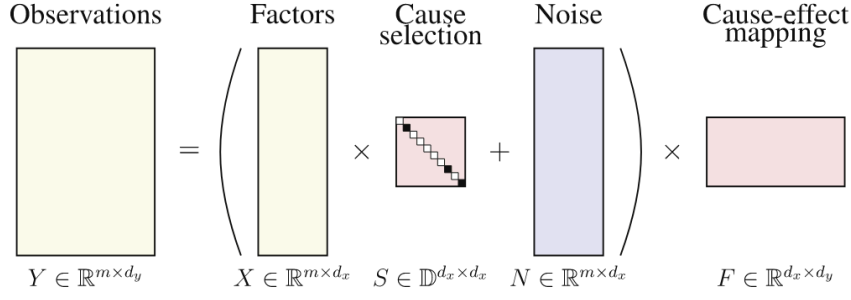
1.2.2. Back-to-Back regression

B2B consists of a decoding step and an encoding step (Figure 2), typically using regularised linear solvers (e.g., ridge regression or a linear SVM). To avoid overfitting, the samples (X, Y) are repeatedly split into disjoint subsets (X_1, Y_1) and (X_2, Y_2) .

In the first step, a backward linear regression model predicts coefficients $\hat{G}$ from multivariate neural observations Y_1 to predictors X_1 . The matrix $\hat{G}$ defines feature-specific supervised readout dimensions in Y -space. The decoded estimates are then used to construct the predictions $\hat{X} = Y_2 \hat{G}$. In the second step, a forward linear regression model uses the original predictors X_2 to predict $\hat{X}$, yielding $\hat{H}$. The diagonal of $\hat{H}$ is the factor-wise, interpretable estimates:

$$\begin{aligned}\hat{G} &= (Y^T Y + \Lambda_Y)^{-1} Y^T X \\ \hat{H} &= (X^T X + \Lambda_X)^{-1} X^T Y \hat{G} \\ \hat{S} &= \text{diag}(\hat{H})\end{aligned}$$

B2B combines both the benefits of forward and backward modelling to provide interpretable and unidimensional coefficient estimates for each factor. B2B has been shown to be a promising analytical method, and it was applied to reading task to decompose different word features that relate to different brain responses. In their results, word length and word frequency emerged as key effects, with evidence consistent with earlier effects in visual processing and later evoked N400 responses.



CC BY-NC-ND 4.0 · King et al. (2020)

$$\begin{aligned}
 & \begin{matrix} X \sim P(X) \\ N \sim P(N) \end{matrix} \quad \hat{S} = \text{diag} \left(\underbrace{(X_2^\top X_2 + \Lambda_X)^{-1} X_2^\top Y_2 (Y_1^\top Y_1 + \Lambda_Y)^{-1} Y_1^\top X_1}_{\substack{1) \hat{X}: \text{regression from } Y \text{ to } X \\ 2) \text{regression from } X \text{ to } \hat{X}}} \right)
 \end{aligned}$$

Figure 2: **B2B algorithm overview.** F denotes an encoding mapping capturing the spatiotemporal EEG pattern associated with each predictor.

2. Planned Project

2.1. Research Question

1. In simulations mimicking co-registration data with known ground truth, does combining overlap correction (rERP) with B2B improve the disentanglement and recovery of collinear predictors under temporal overlap?
2. When applied to ZuCo, does the best-performing pipeline yield stable and interpretable estimates for lexical predictors (surprisal, frequency, length) while controlling for oculomotor covariates?

2.2. Goals

2.2.1. Main Goals

- A. A simulation framework with controllable temporal overlap and predictor collinearity is implemented using Unfold toolbox.
- B. Analysis pipelines spanning the overlap $\times$ collinearity method map (Figure 4), including baseline decoding, rERP, B2B, and rERP + B2B variants, are systematically benchmarked against known ground truth.
- C. A fixation-aware surprisal predictor is implemented, distinguishing first fixations and refixations.
- D. The best-performing pipeline is applied to the ZuCo dataset, including lexical predictors such as word surprisal (from a pretrained language model), log word frequency, and word length, while controlling for oculomotor covariates.

2.2.2. Stretch Goals

- E. Different solver variants in the B2B backward step (L2/ridge vs. linear SVM) are evaluated and compared to assess how solver choice affects decoding accuracy and result stability in the simulations.

2.3. Approach

2.3.1. Overview

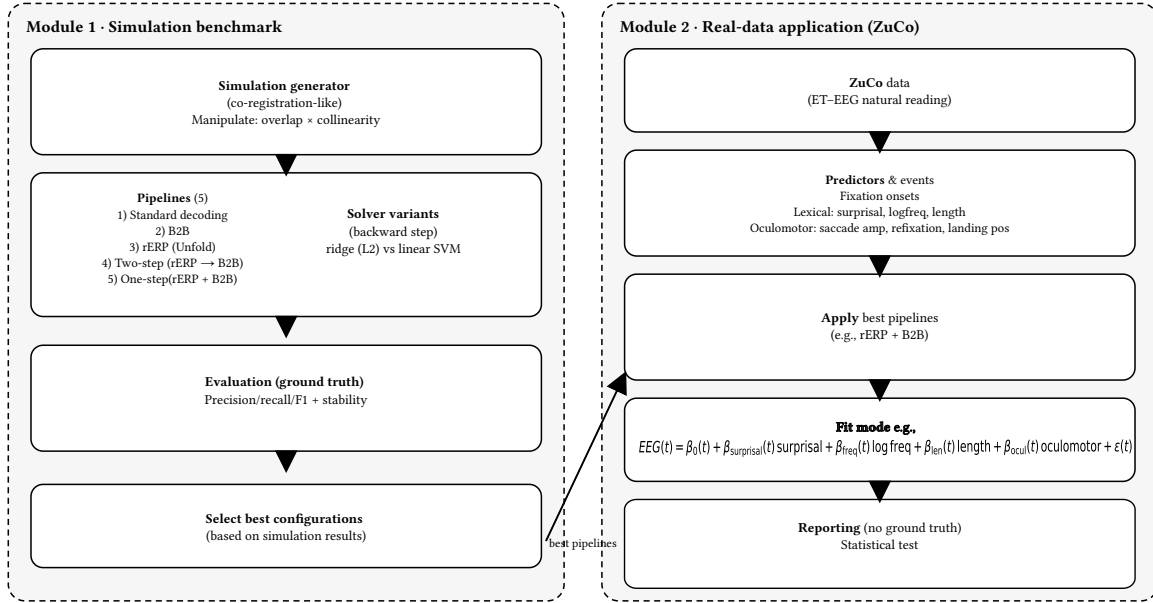


Figure 3: **Project workflow overview:** *Simulation benchmarking* → *select best pipelines* → *ZuCo application*

The project is organised in two stages. First, simulations with known ground truth are used to evaluate whether overlap correction improves B2B’s performance under controlled temporal overlap and predictor collinearity (**Goal A, Goal B, Goal E**). Second, the best-performing pipeline is applied to a real natural reading dataset (ZuCo) as a practical demonstration. (**Goal C, Goal D**)

2.3.2. Simulation design (Goal A)

Synthetic co-registration-like data are generated to mimic fixation-locked responses with controllable overlaps and correlations. Two main factors are manipulated:

- Temporal overlap: none vs. present (implemented by varying event rate / inter-event interval (ISI) distributions).
- Predictor collinearity: none / low / high (implemented by controlling correlations among predictors, using covariate design in Unfold toolbox.)

2.3.3. Analysis pipelines (Goal B)

		Collinearity	
		Low	High
Overlap	None	Standard decoding <i>baseline</i>	Back-to-back regression (B2B) <i>disentangles collinear predictors</i>
	High	Linear deconvolution (rERP) <i>corrects overlap</i>	One-step: joint model (rERP + B2B) Two-step : rERP (single-trial) → B2B

Figure 4: **Method map: overlap x collinearity.**

Five pipelines are compared:

- **Standard decoding:** a baseline decoding model without explicit overlap correction or B2B.
- **Plain B2B:** B2B applied directly to fixation-locked data without explicit overlap correction.
- **Overlap correction only:** deconvolution-based rERP in Unfold without B2B.
- **Two-step overlap-corrected B2B:** overlap correction via deconvolution-based rERP, followed by B2B on overlap-corrected single-trial responses.
- **one-step overlap-corrected B2B:** a joint approach in which overlap correction is incorporated directly into the B2B estimation procedure.

2.3.4. Evaluation criteria (Goal B, Goal E)

- In simulations, ground truth is defined in terms of multi-source predictor-specific waveform shapes; performance is therefore evaluated by the time-course similarity between $\hat{\beta}_i(t)$ and $\beta_i^{\text{true}}(t)$, using e.g. correlation or RMSE. Or cross-talk matrix will be used:

$$M_{ij} = \text{corr}(\hat{w}_{i(t)}, w_j^{\text{true}}(t))$$

Model coefficients: Backward, rERP, B2B, rERP+B2B

Backward sanity check	rERP	B2B	rERP+B2B
$G_i(t) \in \mathbb{R}^{d_y}$	$\hat{\beta}_i(\cdot) \in \mathbb{R}^{d_t}$	$\hat{s}_i(\cdot) \in \mathbb{R}^{d_t}$	$\hat{s}_i^{\text{oc}}(\cdot) \in \mathbb{R}^{d_t}$
Decoding-weights (excluded from waveform-recovery comparisons)	Recovered waveform	Predictor- specific waveform	Overlap-corrected predictor-specific waveform

Figure 5: **Coefficients from different models**

- In ZuCo, where ground truth is unknown, lexical predictor effects will be assessed using knockout-based held-out performance measure:

$$\Delta R_i = R_{\text{full}} - R_{\text{knockout}(i)}$$

Algorithm 2: B2B feature importance.

Input: $X_{train} \in \mathbb{R}^{m \times d_x}$, $X_{test} \in \mathbb{R}^{m' \times d_x}$, $Y_{train} \in \mathbb{R}^{m \times d_y}$, $Y_{test} \in \mathbb{R}^{m' \times d_y}$,
Output: estimate of prediction improvement $\Delta R \in \mathbb{D}^{d_x}$.

```
1  $H, G = \text{B2B}(X_{train}, Y_{train});$   
2  $R_{full} = \text{corr}(X_{test}H, Y_{test}G);$   
3 for  $i = 1, \dots, d_x$  do  
4    $K = Id;$   
5    $K[i] \leftarrow 0;$   
6    $R_k = \text{corr}(X_{test}KH, Y_{test}G_i);$   
7    $\Delta R_i = R_{full} - R_k;$   
8 end  
9 return  $\Delta R$ 
```

CC BY-NC-ND 4.0 · King et al. (2020)

Figure 6: Held-out performance algorithm from B2B paper

2.3.5. Word surprisal (Goal C)

Word surprisal ($-\log p(w_i | w_{<i})$), derived from a pretrained language model as the negative log-probability of each word given its preceding context, will be aligned to the corresponding fixation events in ZuCo and included in the model as lexical predictors. A fixation-aware variant will additionally distinguish first fixations and refixations.

2.3.6. Real-data application: ZuCo (Goal D)

The best-performing configurations are applied to the ZuCo dataset. Fixation-locked events are modelled with Unfold for overlap correction, and B2B is then used to assess factor estimates for lexical predictors (word surprisal, log word frequency and word length) while controlling for oculomotor variables (saccade amplitude, refixation, landing position).

2.3.7. Solver comparison (Goal E)

To assess sensitivity to modelling choices, solver variants in the B2B backward step are compared, focusing on L2/ridge versus linear SVM.

3. Plan

- Review the relevant literature on B2B, FRPs, and deconvolution rERP.
- Implement a simulation benchmark with known ground truth, and evaluate alternative pipelines and solver variants under controlled overlap and collinearity.
- Apply the best-performing configuration to the ZuCo dataset.
- Result consolidation, documentation, and thesis writing.

	Mar	Apr	May	June	Jul	Aug
Literature review, writing proposal	▬					
Simulation + analysis (Goal A-B)		▬	▬	▬		
Zuco application (Goal C-D)				▬	▬	
Buffer / Review / Stretch goals / (Goal E)						▬

Main goal completion

Aug 2026

Bibliography

Degno, F., & Liversedge, S. P. (2020). Eye Movements and Fixation-Related Potentials in Reading: A Review. *Vision*, 4(1), 11. <https://doi.org/10.3390/vision4010011>

- Dimigen, O., Sommer, W., Hohlfeld, A., Jacobs, A. M., & Kliegl, R. (2011). Coregistration of eye movements and EEG in natural reading: Analyses and review. *Journal of Experimental Psychology: General*, 140(4), 552–572. <https://doi.org/10.1037/a0023885>
- Ehinger, B. V., & Dimigen, O. (2019). Unfold: an integrated toolbox for overlap correction, non-linear modeling, and regression-based EEG analysis. *Peerj*, 7, e7838. <https://doi.org/10.7717/peerj.7838>
- Hollenstein, N., Rotsztein, J., Troendle, M., Pedroni, A., Zhang, C., & Langer, N. (2018). ZuCo, a simultaneous EEG and eye-tracking resource for natural sentence reading. *Scientific Data*, 5(1), 180291. <https://doi.org/10.1038/sdata.2018.291>
- King, J.-R., Charton, F., Lopez-Paz, D., & Oquab, M. (2020). Back-to-back regression: Disentangling the influence of correlated factors from multivariate observations. *Neuroimage*, 220, 117028. <https://doi.org/10.1016/j.neuroimage.2020.117028>